Learning Periodic Patterns and Developing Internal Models for Avian-Inspired Flapping-Wing State Estimation

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***Abstract*—This article presents a novel learning-based approach for online state estimation in flapping-wing aerial vehicles (FWAVs). Leveraging low-cost magnetic, angular rate, and gravity (MARG) sensors, the proposed method effectively mitigates the adverse effects of flapping-induced oscillations that challenge conventional estimation techniques. By employing a divide-and-conquer strategy grounded in cycle-averaged aerodynamics, the framework decouples the slow-varying components from the high-frequency oscillatory components, thereby preserving critical transient behaviors while delivering a smooth internal state representation. The complete oscillatory state of FWAV can be reconstructed based on above two components, leading to substantial improvements in state estimation accuracy. Experimental validations on an avian-inspired FWAV demonstrate that the estimator enhances accuracy and smoothness, even under complex aerodynamic disturbances. These encouraging results highlight the potential of learning algorithms to overcome issues of flapping-wing induced oscillation dynamics.**

***Index Terms*—Aerial robot, attitude estimation, flapping-wing, online learning.**

I. Introduction

AUTONOMOUS flapping flight, owing to its high maneuverability on par with birds and insects, offers promising potentials for tremendous applications [1]. Notably, avian-inspired flapping-wing robots can accommodate additional onboard electronics and offer an extended operational range, thereby attracting increased attention in application-oriented research [2], [3], [4], [5]. In order to enhance the survivability of flapping-wing aerial vehicle (FWAV) in complex environments, it is imperative that their attitude estimation, as well as state prediction capabilities, are optimized for autonomous long-distance flight missions [6], [7].

The significance of attitude estimation lies in its ability to integrate noisy and constrained sensor data to produce a precise, real-time comprehension of the swift and unstable motions of an FWAV, thus facilitating safe and autonomous flight. Although attitude estimation is vital for flapping-wing autonomous flight, the community focus more on mechanisms and control techniques, and commonly borrow estimation methods used in conventional robotic systems. These off-the-shelf methods range from extended Kalman filter (EKF) [8], Mahony complementary filter [9] to gradient-based Madgwick filter [10]. According to empirical knowledge, these estimation algorithms can minimally handle flapping-wing flight with low accuracy requirement. Nevertheless, researchers pointed out that the notorious severe vibrations, especially in accelerometer measurement can adversely affect the overall performance of algorithms in specific applications [11], [12]. In early Delfly II autonomous flight research [12], a wingbeat-synchronized moving average filter was used to suppress accelerometer oscillations and estimate gravity for attitude determination. Later work [13] fused high-rate inertial measurement units (IMUs) data (512 Hz) with low-rate optical tracking (120 Hz) via an EKF, enabling sub-cycle aerodynamic analysis without cycle-averaging. These studies reflect two contrasting strategies for handling flapping-induced

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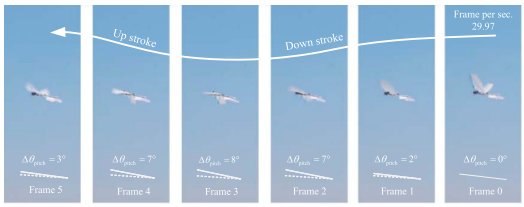


Fig. 1. FWAV in motion: wing flapping yields a nonnegligible periodic pitch motion.

periodic patterns: to attenuate these oscillations or not to attenuate.

In avian-inspired robots with wingspans above 1.0 m, the wing flapping frequency typically remains below 10 Hz, whereas IMUs can sample at rates exceeding 500 Hz. Sufficiently high sampling rate enables the onboard IMU to capture the periodic wing motions, as demonstrated in [14], even with low-cost magnetic, angular rate, and gravity (MARG) sensors. Although these sensors are usually mounted on the fuselage, they effectively register wing movements due to the transmitted fuselage motion (see Fig. 1). According to [15], the adaptive notch filter (ANF) can function as a selective filtering mechanism that isolates the meaningful system dynamics. In [11], a model-based approach estimates oscillatory aerodynamic forces, positing that removing these forces from accelerometer data yields a more accurate gravitational acceleration estimate, however, we contend that the elimination should target only the specific force. Given the complex aerodynamic interactions, extracting periodic patterns via learning methods [16], [17] is preferable over using stroke-angle models, particularly when such measurements are unavailable. In [18], the authors use phase portraits of the wing flap and wing pitch joints to represent different flight maneuvers. An insight derived from these researches suggests that, through learning and eliminating periodic patterns, slowly varying dynamics can be effectively reconstructed, thereby avoiding delays associated with cycle-averaging or conventional filter-based methods.

A critical component for achieving autonomous flapping-wing flight is the attitude control, which necessitates the production of specified wrenches. In the majority of relevant research, a straightforward and effective idea prevails across different scale flapping-wing robots [5], [19], [20], [21]. This idea is to develop a map or a model which can translate the desired wrenches into specific wings and tail settings in a single flapping-wing period [4], [5], [13]. This mapping generates parameters for the periodic movement, and these parameters should be preserved or only minimally adjusted over the duration of one entire flapping cycle. Simplified complexity often contributes to more stable control within FWAV research [22]. Sudden aerodynamic perturbations exacerbate discrepancies between cycle-averaged wrenches and their expected values. From an aerodynamic perspective, it is also crucial to eliminate fluctuating control inputs that may disrupt the airflow surrounding the FWAV, thereby impairing desired wrench generations. The intuitive idea is to use cycle-averaged measurements as feedback signals. However, the frequency of flapping in avian flight, like the developed FWAV shown in Fig. 1, is around 5 Hz. The latest-cycle averaging based attitude estimation, with a delay of approximately 0.1 s (half time of a cycle), is inadequate for achieving agile maneuver.

Modeling the dynamics is a crucial element for attaining online real-time planning and control. The quasi-steady aerodynamics are collectively summarized in Sane and Dickinson's [23] work, which includes added mass effect, translational forces, rotational forces, and the wake-capture mechanisms. Complementing the ideal map discussed earlier, the cycle-averaged model of quasi-steady aerodynamics is widely employed in application-focused research [24], [25], [26], [27]. In their works, the wrenches are averaged over the extent of the wings, and assuming the fuselage oscillation plays a negligible role in the computation. However, wings flapping at approximately 5 Hz can substantially impact the dynamics of the fuselage. In addition, the IMU measurement is mounted on the fuselage, where significant oscillations are also observed.

Building on the previous analysis, this work aims to develop a method for generating prompt, smooth measurements insulated from flapping-induced disturbances. These measurements enable reliable identification of parameters in an internal model, reflecting cognitive-level processing rather than low-level coarse perception. Combined with appropriate estimation algorithms, the internal model yields accurate internal state estimates while still preserving fast dynamics for faithful reconstruction of true oscillatory states.

In accordance with the objectives established in the motivation, the main contributions of this study are encapsulated in the following three aspects.

1) We propose a novel learning-based method that can online estimate the flapping-induced oscillations in the proprioceptive sensing with low-cost MARG sensors.
2) An internal model considering whole-body oscillations is developed based on cycle average, which captures the slow-varying behavior of the avian-inspired FWAV.
3) By employing the divide-and-conquer approach to separately handle the slow-varying and oscillatory dynamics, and subsequently integrating these two components, an accurate comprehensive state estimation is achieved.

The remainder of this article is organized as follows. In Section II, we demonstrate the proposed online periodic pattern learning methods and the smooth state estimation method. The internal model is developed in Section III based on the cycle-averaged aerodynamics, as well as body oscillation information. Then in Section IV, the proposed algorithms and model are validated in real flight experiment of our avian-inspired FWAV. Finally, we conclude this work in Section V.

II. Periodic Gaussian Process Learning

Avian decouples head motion from flapping oscillations and use nystagmus to stabilize vestibular inputs, maintaining equilibrium in flight [28]. Inspired by this, we propose a machine learning approach based on periodic Gaussian processes that autonomously learns and compensates for flapping disturbances in real time. By online periodic learning, the method extracts

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periodic components from acceleration and angular velocity signals, enabling smooth attitude estimation.

A. Periodic Gaussian Process for Pattern Learning

Gaussian process regression (GPR) is a nonparametric approach used in machine learning for regression tasks. It models the function generating the data as a sample from a Gaussian process. This method is especially advantageous when the functional form is not known, and our aim is to deduce the periodic pattern from online flight data.

The GPR learning process aims to achieve that, given a phase value $\phi \in [0, 2\pi)$, the regression can estimate the oscillatory component of certain measurement. It takes as input the collection of observed learning phases $\boldsymbol{\Phi} = [\phi_1\ \phi_2\ \dots\ \phi_n]^\top \in [0, 2\pi)^n$ and as output the corresponding oscillatory measurements $\boldsymbol{Y} = [Y_1\ Y_2\ \dots\ Y_n]^\top \in \mathbb{R}^n$.

With the purpose of removing the nonzero prior mean from the periodic pattern, the following operation is implemented before feeding the data into GPR process:

$$\boldsymbol{Y}_{\text{osc}} = \boldsymbol{Y} - \frac{\sum_{i=1}^{\dim(\boldsymbol{Y})} Y_i}{\dim(\boldsymbol{Y})} \tag{1}$$

where $\boldsymbol{Y}_{\text{osc}}$ is the actually applied output observation, Y_i represents the component in the vector $\boldsymbol{Y}$, and $\dim(\boldsymbol{Y})$ is the dimension of the output vector.

Given that the function we are modeling is recognized as nearly periodic, it is natural to use Fourier basis functions as the kernel to model periodicity [29]. Implemented as the feature vector function, the truncated Fourier basis is

$$\boldsymbol{F}(\phi) = [\sin(\phi)\ \cos(\phi)\ \cdots\ \sin(n_\phi\phi)\ \cos(n_\phi\phi)]^\top \tag{2}$$

where ϕ represents the periodic coordinate in a cycle progression, and $n_\phi \in \mathbb{N}^+$ is the highest harmonics of the phase. Increasing n_ϕ can produce richer but less smooth periodic functions. Subsequently, we define the feature space spanned by $\boldsymbol{F}$ as $\mathcal{H}$. The reproducing kernel can be given as

$$k_p(\phi_a, \phi_b) = \boldsymbol{F}^\top(\phi_a) G^{-1} \boldsymbol{F}(\phi_b) \tag{3}$$

where ϕ_a and ϕ_b are two different phases, and G is the gram matrix whose elements are shown as $G_{i,j} = \langle F_i, F_j \rangle_{\mathcal{H}}$. The inner product with respect to the space $\mathcal{H}$ is given by

$$\langle F_i, F_j \rangle_{\mathcal{H}} = \frac{1}{2\pi} \int_0^{2\pi} F_i(\phi) F_j(\phi) \mathrm{d}\phi \tag{4}$$

where F_i represents the ith element of the feature vector function $\boldsymbol{F}$. Thus, $G = (1/2)\, I_{2n_\phi} \in \mathbb{R}^{2n_\phi \times 2n_\phi}$. Consequently, the reproducing kernel given by (3) can be reformulated as

$$k_p(\phi_a, \phi_b) = 2\boldsymbol{F}^\top(\phi_a) \boldsymbol{F}(\phi_b). \tag{5}$$

According to [30], the joint distribution of the observed values and the function values at a new phase of interest obeys

$$\begin{bmatrix} \boldsymbol{Y}_{\text{osc}} \\ f_* \end{bmatrix} \sim \mathcal{N}\left(\boldsymbol{0}, \begin{bmatrix} K_p(\boldsymbol{\Phi}, \boldsymbol{\Phi}) + \sigma_n^2 I & \boldsymbol{K}_p(\boldsymbol{\Phi}, \phi_*) \\ \boldsymbol{K}_p(\boldsymbol{\Phi}, \phi_*)^\top & k_p(\phi_*, \phi_*) \end{bmatrix}\right) \tag{6}$$

where $\boldsymbol{\Phi}$ is the input vector, $\boldsymbol{Y}_{\text{osc}}$ is the output observation vector, $\phi_\star \in (0, 2\pi]$ is the new phase of interest, and $f_\star \in \mathbb{R}$ represents the oscillation prediction. The positive variable $\sigma_n \in \mathbb{R}^+$ represents the observation noises. The covariance matrix can be constructed through the following manner:

$$K_{p,ij} = k_p(\phi_i, \phi_j) \tag{7}$$

where $K_{p,ij} \in \mathbb{R}$ is the ith row and jth column of the matrix K_p. Through investigating the conditional distribution, we can obtain the predictive equations for: GPR

$$\begin{aligned} \bar{f}_* &\triangleq \mathbb{E}\left[f_* \mid \boldsymbol{\Phi}, \boldsymbol{Y}, \phi_\star\right] \\ &= \boldsymbol{K}_p^\top(\boldsymbol{\Phi}, \phi_\star) \left[K_p(\boldsymbol{\Phi}, \boldsymbol{\Phi}) + \sigma_n^2 \boldsymbol{I}\right]^{-1} \boldsymbol{Y}_{\text{osc}} \end{aligned} \tag{8a}$$

$$\begin{aligned} \text{var}\left(f_*\right) &= k_p(\phi_\star, \phi_\star) \\ &\quad - \boldsymbol{K}_p^\top(\boldsymbol{\Phi}, \phi_\star) \left[K_p(\boldsymbol{\Phi}, \boldsymbol{\Phi}) + \sigma_n^2 \boldsymbol{I}\right]^{-1} \boldsymbol{K}_p(\boldsymbol{\Phi}, \phi_\star) \end{aligned} \tag{8b}$$

where $\bar{f}_* \in \mathbb{R}$ and $\text{var}\left(f_*\right) \in \mathbb{R}$ are the estimation of the mean value and the variance, respectively. Implementing these formulae, we can derive the output estimation and its associated uncertainty for an arbitrary phase of interest. It should be noted that, since $\boldsymbol{\Phi}$ is a vector whereas $\phi_\star$ is scalar, the term $\boldsymbol{K}_p^\top(\boldsymbol{\Phi}, \phi_\star)$ is thus a vector, and denoted in bold. Simultaneously, by (7), $K_p(\boldsymbol{\Phi}, \boldsymbol{\Phi})$ is still a matrix.

Remark 1: The learning phase ϕ should not be interpreted as a direct description of wing motion. Instead, it serves as an abstract coordinate within the periodic pattern. The map from the phase ϕ to oscillatory measurement is in the order of cyclic progression, but self-organized by the regression process, allowing the model to capture periodic patterns without a regimented correspondence to kinematic states.

As illustrated in Fig. 2, periodic patterns of signals, notably the Z-component of acceleration and the Y-component of angular velocity, are effectively captured by periodic-GPR, which maintains uniformity across all phases, producing smooth learning results for further estimations.

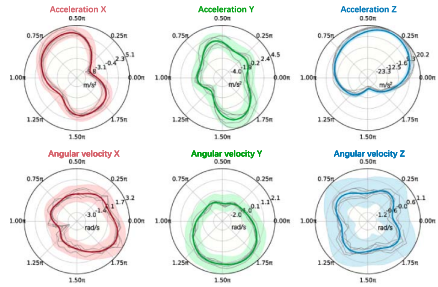


Fig. 2. Polar plots demonstrating the learning of periodic sensing patterns via GPR. To reduce sampling bias, data were acquired in integer cycles. The original, unprocessed sensor measurements used to train the periodic GPR model are displayed as gray traces, repeating across the four cycles. All data were collected in real flights. Darker-colored curves indicate the posterior mean predictions, while the lighter shaded regions correspond to the 95% confidence intervals, approximated as ±2 standard deviations around the mean.

B. Period Length Estimation

The Cooley–Tukey fast Fourier transform (FFT) algorithm with $\mathcal{O}(N \log N)$ complexity efficiently computes the discrete

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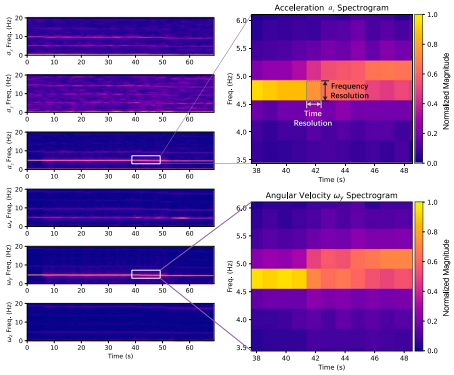


Fig. 3. Spectrograms over time generated by sliding-window Cooley–Tukey FFT algorithm with real flight data. The signals of acceleration and angular velocity are employed for frequency estimation due to their superior update rates compared to all other available signals.

Fourier transform (DFT) by recursively decomposing it into smaller DFTs using a *divide-and-conquer* strategy [31]. In situations where the FWAV experiences significant interference causing sensor data to be collected in a sparse manner, the Lomb–Scargle periodogram [32], [33] can be employed.

The sliding-window Cooley–Tukey FFT results from real-flight data are shown in Fig. 3. The frequency data can be extracted from the Z-component of acceleration a_z and the Y-component of angular velocity ω_y signals. The right sub-figures demonstrate that the frequencies extracted from both signals are closely aligned, indicating that the frequency extraction process is relatively stable. In DFT analysis, extending the observation window improves frequency resolution but reduces time resolution, limiting the capture of rapid temporal changes. Conversely, increasing time resolution lowers frequency resolution, making it harder to separate closely spaced frequencies. At a given time instant, modeling the frequency with a Gaussian distribution, characterized by mean $\bar{f}_{\text{freq}} \in \mathbb{R}^+$ and variance $\sigma^2_{\text{freq}} \in \mathbb{R}^+$, provides a compact evaluation by integrating frequency estimates from a_z and ω_y.

C. Signals Smoothing by k-Means on Unit Sphere

Sphere k-means is particularly effective for managing phase-based data, such as angular and directional data, since it operates under the premise that data points are situated on the surface of a unit sphere [34].

In this context, the cosine similarity can be expressed as

$$d_{\cos}(\phi_1, \phi_2) = 1 - \cos\left(\frac{\phi_1 - \phi_2}{2}\right) \tag{9}$$

where ϕ_1 and ϕ_2 are two distinct phase angles. The k-means cluster algorithm can be transformed into the optimization problem $\min_{C_\phi} \mathcal{C}(C_\phi)$. In order to minimize the total distance (9) between each data phase and its closest cluster centroid, the optimization objective function is given by

$$\mathcal{C}(C_\phi) = \sum_{\phi_d \in D_\phi} \sum_{\phi_c \in C_\phi} \delta(l_{C_\phi}(\phi_d), \phi_c) d_{\cos}(\phi_d, \phi_c) \tag{10a}$$

$$l_{C_\phi}(\phi) = \arg\min_{\phi_c \in C_\phi} d_{\cos}(\phi, \phi_c) \tag{10b}$$

$$\delta(\phi_1, \phi_2) = \begin{cases} 1 & \text{if } \phi_1 = \phi_2 \\ 0 & \text{if } \phi_1 \neq \phi_2 \end{cases} \tag{10c}$$

where D_ϕ is the set of phases corresponding to the overall data stored in memory, and C_ϕ is the set of cluster centroids based on the data in memory. The function $l_{C_\phi}(\phi)$ identifies the closest cluster centroid relative to ϕ. The Kronecker delta function δ is employed to constrain the distance accumulation within each category. To enhance computational efficiency, optimization-based k-means is used only during initialization, and subsequent updates rely on the more efficient traditional k-means, which is invoked only when kernel aggregation becomes significant.

After k-means clustering, each data point is assigned a cluster label. Evaluation is performed by averaging values within each cluster, producing both mean signals and phase angles for periodic-GPR learning, respectively corresponding to output observation and observed input. The observation variance per cluster is also straightforwardly computed and used as σ_n^2 in (8a). As shown in Fig. 2, the use of only 14 cluster centers provides sufficient representational capacity, such that the uncertainties underlying are primarily due to observation noise rather than distances to cluster centers.

D. Periodic Learning and Estimation Pipeline

The aforementioned algorithm necessitates an initial phase, indicated as ϕ_0. Utilizing a single-pole low-pass filter, the filtered phase changing rate ω_{flt} can be obtain by input the instantaneous phase changing rate, which is given by $2\pi\bar{f}_{\text{freq}}$. The phase angle ϕ is then updated by

$$\phi(t_{\text{curr.}}) = \phi(t_{\text{last}}) + \omega_{\text{flt}}(t_{\text{curr.}} - t_{\text{last}}) \tag{11a}$$

$$\phi(t_0) = \phi_0 \tag{11b}$$

where $t_{\text{curr.}} \in \mathbb{R}$ is the current time instant, and $t_{\text{last}} \in \mathbb{R}$ is the preceding time instant. In order to make periodic learning phase-aligned, phase correction is utilized and accomplished through the application of cross-correlation

$$R_{\text{cc}}(k) = \sum_{i=-\lceil f_s/\bar{f}_{\text{freq}} \rceil}^{\lceil f_s/\bar{f}_{\text{freq}} \rceil} \boldsymbol{s}(i)^\top \hat{\boldsymbol{s}}(i+k) \tag{12a}$$

$$\phi_{\text{d}} = \frac{2\pi\bar{f}_{\text{freq}}}{f_s} \underset{k \in \{-\lceil f_s/\bar{f}_{\text{freq}} \rceil \cdots \lceil f_s/\bar{f}_{\text{freq}} \rceil\}}{\arg\max} R_{\text{cc}}(k) \tag{12b}$$

$$\phi'(t_{\text{curr.}}) = \phi(t_{\text{curr.}}) + k_{\text{cc}} \min(\max(\phi_{\text{d}}, \underline{\phi}_{\text{d}}), \overline{\phi}_{\text{d}}) \tag{12c}$$

where $R_{\text{cc}}(k) \in \mathbb{Z} \to \mathbb{R}$ is the cross-correlation with difference of k, with $k \in \mathbb{Z}$ being the data sequence number, $\lceil \star \rceil \in \mathbb{R} \to \mathbb{Z}$ rounds up $\star$ to the nearest integer, the selected signal vector $\boldsymbol{s} = [a_z\ \omega_y]^\top \in \mathbb{R}^2$ collects the signals with distinctive frequencies, meanwhile, $\hat{\boldsymbol{s}} \in \mathbb{R}^2$ is its corresponding estimation from GPR. Moreover, $f_s \in \mathbb{R}$ is the sampling frequency of the sensors. Furthermore, $\phi_{\text{d}} \in \mathbb{R}$ signifies the phase difference. Concurrently, $\underline{\phi}_{\text{d}} \in \mathbb{R}^-$ and $\overline{\phi}_{\text{d}} \in \mathbb{R}^+$ indicate its lower and upper bounds, respectively. And the constant $k_{\text{cc}} \in \mathbb{R}^+$ is the correction update rate. Finally, we obtain the new phase angle estimation $\phi' \in (0, 2\pi]$.

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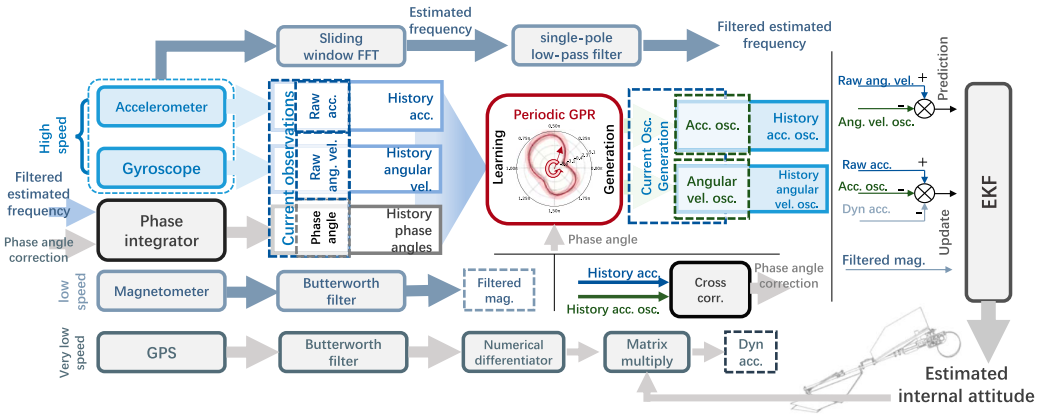


Fig. 4. Framework for implementing periodic Gaussian process learning in the estimation of internal attitude states.

We use periodic acceleration learning as an illustrative example. Since the phase estimation (11) and (12) can be obtained at any time instance, a data tuple as $\{\boldsymbol{a}_\star, \phi_\star\}$ becomes available immediately upon receiving acceleration data. These data tuples are then stored in a memory stack, wherein older entries are substituted by new ones once the stack attains its full capacity. Following this, the data are selected by k-means as demonstrated in (10), and the obtained cluster centers are utilized in periodic GPR learning.

The comprehensive state estimation method is illustrated in Fig. 4. The rapidly updated raw data displaying oscillations from the flapping-wing movement, specifically the acceleration and angular velocity after subtraction of their respective centered averages, serve as inputs for periodic GPR learning, alongside the phase estimation results. Upon provision of a phase angle, the periodic GPR process outputs the respective oscillation components, which are consistently estimated in real-time. The oscillation-free signals are subsequently processed by an EKF to estimate attitude unaffected by flapping-wing-induced oscillations.

With the raw acceleration observation $\boldsymbol{a}$ and the estimated acceleration oscillation $\hat{\boldsymbol{a}}_{\text{osc}}$ in hand, the internal acceleration signal can be obtained as $\boldsymbol{a} - \hat{\boldsymbol{a}}_{\text{osc}}$. The angular velocity data undergo the same operation, whereas the geomagnetic vector data and global positioning system (GPS) data do not, since their slow variation and lower sampling rate make them less susceptible to oscillatory disturbances. The dynamic acceleration can be estimated with feedback from the GPS. The gravitational acceleration is then determined by subtracting the dynamic acceleration from the total acceleration measurement. Finally, the internal signals are processed using the standard EKF method [35], enabling the estimation of internal attitudes.

III. Internal Translational Model Development

An internal model in the sensorimotor system is a neural mechanism that predicts the sensory outcomes of actions to enable accurate and adaptive motor control [36]. Accordingly, we derive internal aerodynamic coefficients in sense of cycle average from the instantaneous aerodynamic coefficients, considering oscillations on whole body. To this end, we develop

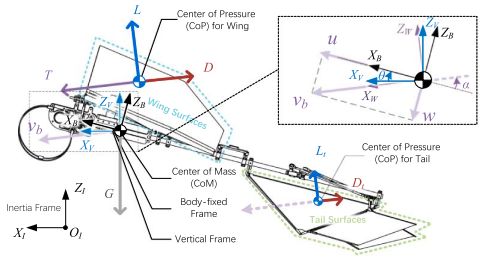


Fig. 5. Longitudinal reference framework of the FWAV along with forces exerted upon it. The longitudinal inertia frame is denoted by X_I-Z_I, whereas the longitudinal body-fixed frame is represented by X_B-Z_B. The wind frame, X_W–Z_W, is defined based on the translational velocity. The deflection angles of the ruddervators on the inverted-V tail are anticipated to cause minimal variation in lift on the tail surfaces.

an internal model to describe the translational dynamics of the FWAV. The derivation primarily addresses longitudinal dynamics, whereas the 3-D translational model can be determined by integrating cycle-averaged aerodynamic coefficients.

A. Instantaneous Aerodynamics

Since periodic aerodynamic forces mainly acts on the longitudinal plane of the FWAV, we mainly discuss the longitudinal dynamics subsequently. The related frames and aerodynamic forces during flight are clearly illustrated in Fig. 5. Body velocity relative to wind $V_b \in \mathbb{R}$ along X_B, Y_B and Z_B are denoted as u, v and $w \in \mathbb{R}$, respectively. Meanwhile, $\alpha \in \mathbb{R}$ is the angle of attack (AoA) defined as $\alpha = \tan^{-1}(-w/u)$ while $\beta \in \mathbb{R}$ indicates the side slip angle defined as $\beta = \sin^{-1}(-v/V_b)$. Distinct from the wing AoA, the aforementioned AoA is defined using the fuselage as the reference body, due to its ease of observation and computational efficiency. The pitch angle, $\theta \in \mathbb{R}$, and the pitch angular rate, $q \in \mathbb{R}$, are kinematically related through

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = q. \tag{13}$$

The movement of the reference wing chord in the circumferential direction can be described by the following time-depended function $h(t) = h_0 \cos(2\pi f t)$, where $f \in \mathbb{R}^+$ represents the frequency of flapping and $h_0 \in \mathbb{R}^+$ denotes

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the amplitude of the translational movement. The reference wing chord refers to the chord located at 1/3 of the semi-span distance from the wing tip [26]. The stroke motion phase $\psi \in \mathbb{R}$ can be governed by the function $\psi(t) = 2\pi ft$. It is assumed that the angular motions associated with the flapping wing can be expressed as sinusoidal functions of ψ. Let $L \in \mathbb{R}$, $T \in \mathbb{R}^+$, and $D \in \mathbb{R}^+$ represent the lift, thrust, and drag from the wing, respectively. Meanwhile, $L_t \in \mathbb{R}$ and $D_t \in \mathbb{R}^+$ denote the lift and drag from tail, respectively. Finally, the gravitational acceleration magnitude is given as $g \in \mathbb{R}^+$.

The FWAV has a flapping amplitude $h_0 \approx 0.2\,\mathrm{m}$. Reynolds number during flapping flight is approximately 2×10^5 and Strouhal number 0.15, within the ranges where the Fernandez-Feria's theory is effective [37] to model thrust. We use instantaneous aerodynamic coefficients presented in [38], which are corrected on the basis of [39]. Moreover, in [26], using these coefficients leads to high precision in dynamic modeling under flight conditions similar to ours. Importantly, the authors highlight that these coefficients can still offer a satisfactory approximation, even though the flapping amplitude slightly exceeds the applicability of linear aerodynamic models, and the effect of wing flexibility is negligible.

Aerodynamic characteristics can be determined by critical aerodynamic coefficients denoted as $C_\star$, where $\star$ can correspond to L, T, D and $\star_t$ correspond to L_t, D_t. The coefficient can be identified through $C_\star = (\star + \star_t)/QS$, where $Q \in \mathbb{R}^+$ denotes the dynamic pressure, and $S \in \mathbb{R}^+$ represents the wing area. As stated by [23], lift is comprised of translation force, rotational force, wake capture force, and inertia force. Given that they oscillate with the same period, their representations are simplified by introducing the phase delay between the lift oscillation and stroke angle, denoted as $\psi_L \in \mathbb{R}$. The amplitude is indicated as $C_{L,\mathrm{osc}} \in \mathbb{R}^+$. Thus, the lift coefficient $C_L \in \mathbb{R}$ is shown as follows:

$$C_L = C_{L\alpha}\alpha + C_{L,\mathrm{osc}} \sin(\psi + \psi_L) \tag{14}$$

where $C_{L\alpha} \in \mathbb{R}^+$ is lift curve slope, and the linear term $C_{L\alpha}\alpha$ is the simplified quasi-steady lift coefficient under small AoA assumption. According to [40], lift coefficient of thin airfoil is proportional to $\sin(2\alpha)$. The simplified coefficient can be viewed as the first-order approximation and results in an error less than 10% as long as $\alpha < 0.4$. Meanwhile, due to small area and low aspect ratio of the tail, we neglect L_t, which introduces approximately an 11% error to lift calculation. We define cycle-averaged AoA as $\bar{\alpha} \in \mathbb{R}$ and the periodic oscillation of AoA as $\tilde{\alpha}_{\mathrm{osc}} = \alpha_{\mathrm{osc}} \sin(\psi + \psi_\alpha)$, where $\alpha_{\mathrm{osc}} \in \mathbb{R}^+$ is oscillation amplitude of AoA and $\psi_\alpha \in \mathbb{R}$ is phase delay between AoA oscillation and stroke angle. Hence, the thrust coefficient $C_T \in \mathbb{R}$ obeys

$$\begin{aligned} C_T = {} & C_{Th}(kh_0)^2 \sin(2\pi ft)[F_1(k)\cos(2\pi ft) \\ & - G_1(k)\sin(2\pi ft)] - (\bar{\alpha} + \tilde{\alpha}_{\mathrm{osc}})C_L \end{aligned} \tag{15}$$

where $C_{Th} \in \mathbb{R}^+$ represents a proportional coefficient affected by wing aspect ratio, $k \in \mathbb{R}^+$, $G_1(k) \in \mathbb{R}^-$ and $F_1(k) \in \mathbb{R}^-$ are respectively reduced frequency, the imaginary and real parts of the Theodorsen's function $C_1(k)$ defined in [39].

Drag coefficient $C_D \in \mathbb{R}^+$ contains zero-lift drag coefficient $C_{D0} \in \mathbb{R}^+$ and induced drag coefficient $C_{Di} \in \mathbb{R}^+$. The symbol C_{D0} corresponds to the friction and manifests as a constant, meanwhile, C_{Di} can be computed based on C_L, which is also inversely proportional to aspect ratio of the wing. The inversely proportional coefficient is denoted as $A \in \mathbb{R}$. The impact of the induced drag produced by the tail is represented by a coefficient $k_t \in \mathbb{R}$ that exceeds 1. In conclusion, the drag coefficient C_D follows:

$$C_D = C_{D0} + C_{Di} = C_{D0} + \frac{k_t C_L^2}{A}. \tag{16}$$

B. Internal Translational Model

In this section, we determine the internal aerodynamic coefficients by averaging the magnitude of the aerodynamic coefficients over a flapping period, which is defined in the cycle-averaged wind frame

$$\bar{C}_L = \frac{1}{2\pi}\int_0^{2\pi} (C_L \cos\tilde{\theta}_{V,\mathrm{osc}} - (C_T - C_D)\sin\tilde{\theta}_{V,\mathrm{osc}})\mathrm{d}\psi \tag{17a}$$

$$\bar{C}_T = \frac{1}{2\pi}\int_0^{2\pi} (C_T \cos\tilde{\theta}_{V,\mathrm{osc}} + C_L \sin\tilde{\theta}_{V,\mathrm{osc}})\mathrm{d}\psi \tag{17b}$$

$$\bar{C}_D = \frac{1}{2\pi}\int_0^{2\pi} C_D \cos\tilde{\theta}_{V,\mathrm{osc}}\mathrm{d}\psi \tag{17c}$$

where $\tilde{\theta}_{V,\mathrm{osc}} = \tilde{\alpha}_{\mathrm{osc}} + \tilde{\theta}_{\mathrm{osc}}$ denotes the oscillating velocity-direction in $X_B - Z_B$ plane, where $\tilde{\theta}_{\mathrm{osc}}$ is oscillating pitch angle. Note that due to body oscillation during flight, we cannot straightforwardly average (14)–(16) as internal aerodynamic coefficients of whole body. The oscillation of angular velocity along Y_B is $\tilde{q}_{\mathrm{osc}} = q_{\mathrm{osc}} \sin(\psi + \psi_q)$, where $q_{\mathrm{osc}} \in \mathbb{R}^+$ represents the oscillation amplitude, and $\psi_q \in \mathbb{R}$ is the difference between oscillation phase and the stroke phase ψ.

Due to $\tilde{\theta}_{V,\mathrm{osc}}$ is small, approximately 0.15 during flight, $\sin\tilde{\theta}_{V,\mathrm{osc}} \approx \tilde{\theta}_{V,\mathrm{osc}}$. Substitute (14)–(16), and (13) into (17a)–(17c), internal aerodynamic coefficients is

$$\bar{C}_L = C_{L\alpha}\bar{\alpha} \tag{18a}$$

$$\begin{aligned} \bar{C}_T = {} & -\frac{1}{2}C_{Th}(kh_0)^2 G_1(k) - \bar{\alpha}\bar{C}_L \\ & - \frac{C_{L,\mathrm{osc}} q_{\mathrm{osc}} \sin(\psi_q - \psi_L)}{4\pi f} \end{aligned} \tag{18b}$$

$$\bar{C}_D = C_{D0} + \frac{1}{A}(k_t\bar{C}_L^2 + \frac{1}{2}C_{L,\mathrm{osc}}^2). \tag{18c}$$

During flight, $C_{L,\mathrm{osc}}$, q_{osc}, and phase difference $\psi_q - \psi_L$ are estimated online with the help of the method given in the previous section. The online process maintains sliding sequences of $\hat{a}_{z,\mathrm{osc}}$ and $\hat{\omega}_{y,\mathrm{osc}}$ with length of approximately one flapping period. The maximal absolute value of each sequence is considered as the approximation of the oscillation amplitude, since the oscillation is kept zero-mean. These values are deemed as the required coefficients $C_{L,\mathrm{osc}}$ and q_{osc}, respectively. Subsequently, we identify the value of $\phi_{\mathrm{tran.}}$ at which $\hat{a}_{z,\mathrm{osc}}$ transitions from negative to positive, and then find $\hat{\omega}_{y,\mathrm{osc}}$ value at $\phi_{\mathrm{tran.}}$. Consequently, it follows $\sin(\psi_q - \psi_L) =$

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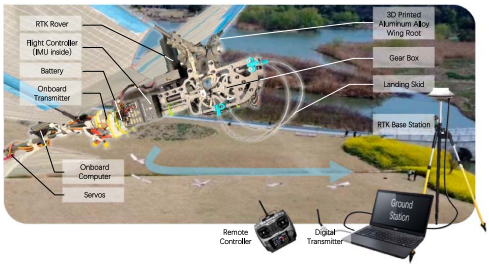


Fig. 6. Real flight experiment settings: the FWAV is remotely controlled, supported by an attitude control realized onboard. High-precision positioning is achieved by fusing real-time kinematic (RTK) rover signals onboard the FWAV with data transmitted from the RTK base station.

$\hat{\omega}_{y,\mathrm{osc}}(\phi_{\mathrm{tran.}})/q_{\mathrm{osc}}$. In contrast, the remaining coefficients: $C_{L\alpha}$, $C_{Th}G_1(k)$, C_{D0}, A and k_t are determined by the robot aerodynamic configuration, which are determined offline according to [26].

Finally, the overall internal translational dynamics model in 3-D space can be represented by

$$\frac{\mathrm{d}\bar{\boldsymbol{p}}}{\mathrm{d}t} = \bar{\boldsymbol{v}} \tag{19a}$$

$$\frac{\mathrm{d}\bar{\boldsymbol{v}}}{\mathrm{d}t} = -g\boldsymbol{e}_3 + R_B^I R_W^B \boldsymbol{f}^W. \tag{19b}$$

The aerodynamic force in the wind frame $\boldsymbol{f}^W \in \mathbb{R}^3$ is expressed as $\boldsymbol{f}^W = QS \cdot \begin{bmatrix}\bar{C}_T - \bar{C}_D & C(\beta) & \bar{C}_L\end{bmatrix}^\top$, where $C(\beta)$ is lateral force coefficient. According to [41] it is proportional to β during steady flight. Let $\bar{\boldsymbol{p}} \in \mathbb{R}^3$ and $\bar{\boldsymbol{v}} \in \mathbb{R}^3$ denote the position and velocity vectors in the internal model, respectively. The matrix $R_B^I \in \mathrm{SO}(3)$ represents the rotation from the body-fixed frame to the inertial frame, while $R_W^B \in \mathrm{SO}(3)$ denotes the rotation matrix from the wind frame to the body-fixed frame. Moreover, $\boldsymbol{e}_3 = [0 \quad 0 \quad 1]^\top$ is the negative gravitation direction in the inertia frame.

IV. Experiments and Analyses

Extensive flight experiments for the FWAV developed in this work were carried out at the Zijingang Campus, Zhejiang University,[1] Huzhou Xishanyang Wetland Park, and High-tech zone Campus, University of Science and Technology of China, respectively. As illustrated in Fig. 6, the FWAV is equipped with a dc brushless motor, an electric speed controller (ESC), a Holybro Pixhawk 6C flight controller, an NVIDIA Jetson Orin NX onboard computer, an 800 mAh 4S LiPo battery, and two servos for controlling the ruddervators. Moreover, the FWAV possesses a wingspan measuring 1.7 m and a mass of 1100 g. The online estimation algorithm is implemented using C++ and operates in conjunction with the robot operating system (ROS) on a Linux Ubuntu platform.

The desired attitude is manually set and fed to the attitude controller, which runs onboard the flight controller. MARG sensor signals with timestamps are sent to the onboard computer, with acceleration and angular velocity updated at 170.8 Hz, magnetic data at 13.4 Hz, and GPS at 8.7 Hz. Second-order Butterworth filters standardize all data to 200 Hz.

[1]See video at https://www.youtube.com/watch?v=$lavNlu5x11Q and codes https://github.com/XJX-UAV/Flapping-wing-State-Estimation

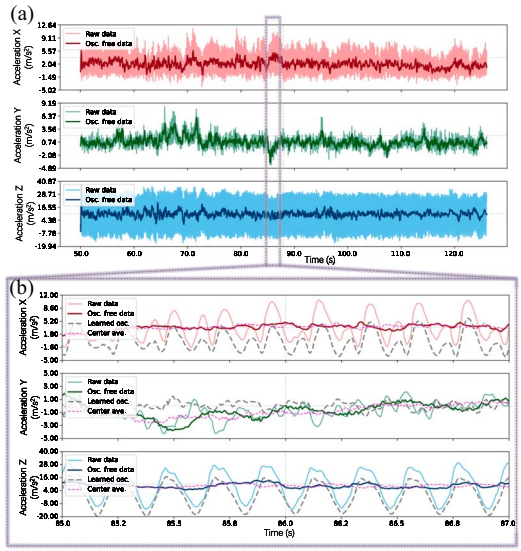


Fig. 7. Acceleration oscillation learning in real flapping-wing flight. "Raw data" refers to the unprocessed measurements directly acquired from the onboard sensors. "Osc. free data" denotes the data minus the oscillation components. "Learned osc." indicates the oscillations learned from the proposed algorithm. And "Center ave." represents centered cycle-averaged data. (a) Raw acceleration signals and oscillation-free data over a 75-second flight segment. (b) A zoomed-in 2-second interval illustrating raw signals, oscillation-free data, learned oscillatory components, and centered cycle-averaged signals.

A. Periodic GPR Learning

The periodic GPR learning algorithm functions in real-time and is implemented onboard. Meanwhile, the FFT algorithm is performed based on the Z-component of acceleration, denoted as a_z, and the Y-component of angular velocity, denoted as ω_y, each with a length of 512 data points, in radix-2 form to optimize computational efficiency [42], while suitably balancing precision and agility. Given that the flapping-wing frequency in cruise flight is around 5.0 Hz, frequencies less than 1.0 Hz or exceeding 8.0 Hz are excluded from the FFT intensity evaluation. The cross-correlation is performed based on one flapping-wing cycle data. Moreover, in the employed k-means algorithm, 14 cluster centers are utilized and are calculated in real-time. The number of centers, serving as GPR kernels, is chosen to ensure that the epistemic uncertainty of GPR converges to an acceptable level, while keeping the number as small as possible to maintain computational efficiency. Given the experimental settings described above, the acceleration results of the online periodic pattern learning are ultimately depicted in Fig. 7. To validate the robustness of the proposed algorithm, experiments corresponding to Fig. 7 are performed under windy conditions, where the maximum wind speed exceeds 5.0 m/s.

As demonstrated in Fig. 7(a), the proposed algorithm effectively remove oscillatory measurements, resulting in signals with reduced oscillations that are more amenable to subsequent internal attitude estimation. In the zoom-in view presented in Fig. 7(b), empirical analysis reveals that, with the application of FFT and cross-correlation corrections, the periodic pattern derived through online learning demonstrates a pronounced alignment with the observable trends in the raw data. For practical purposes, high-cutoff frequency filtering is applied to the

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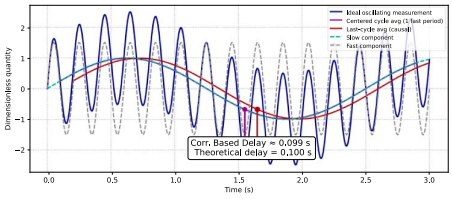


Fig. 8. Conceptual illustration of the time delay due to latest-cycle-averaging: the slow component is shown as $s(t) = \sin(0.4 \cdot 2\pi t)$, whereas the fast component follows $f(t) = 1.5\sin(5.0 \cdot 2\pi t)$. The theoretical delay of the latest-cycle average is half of the fast component period, 0.1 s. The "corr. based delay" is computed by cross-correlation.

raw observation values minus the learned oscillation patterns, resulting in acceleration oscillation-free data. Moreover, it can be observed that the oscillation-free acceleration data remains in close proximity to the centered cycle-averaged data via a sliding window with a length equivalent to one flapping cycle. The centered cycle-averaged data serves as an approximation of the ground truth. However, its online computation is impractical because future data are not available.

Remark 2: Conventional flapping-wing attitude estimations are usually based on cycle-averaging [12], [43]. As illustrated in Fig. 8, the centered cycle-average recovers the underlying slow trend of an ideal oscillatory measurement composed of both fast and slow components. However, this approach requires access to future samples, which is infeasible in real-time operation. Consequently, in online applications the averaging can only be performed over the latest completed cycle, unavoidably introducing a time delay.

Based on the curves illustrated in Fig. 9(a), the algorithm consistently attenuates the periodic oscillations in all three dimensions. According to the learning performance shown in Fig. 9(b), when a temporary side wind is exerted on the FWAV, the roll angular velocity drastically changes, during which the learning process is unable to adequately adapt. Consequently, the oscillation-free data cease to exhibit flawless smoothness in roll rotation. The motion corresponding to Fig. 9(c) also suffers a drastic roll rotation. Nonetheless, in this scenario, the absence of lateral wind allows the periodic pattern to maintain its current configuration, thereby allowing the oscillation-free data to maintain the relative smoothness. In conclusion, the algorithm leverages recent observations to build its periodic pattern recognition, allowing it to adapt to gradual changes while exhibiting sensitiveness to abrupt disturbances. Notwithstanding that averaging over the last cycle can be computed online, this approach results in a half-cycle delay when compared to centered cycle-averaging, as indicated by the delay measure and analysis in boxes, which can be found in Fig. 9(c). Conversely, oscillation-free data from the proposed algorithm can be executed online and exhibit no obvious delay (correlation-based delay as 0.04 s).

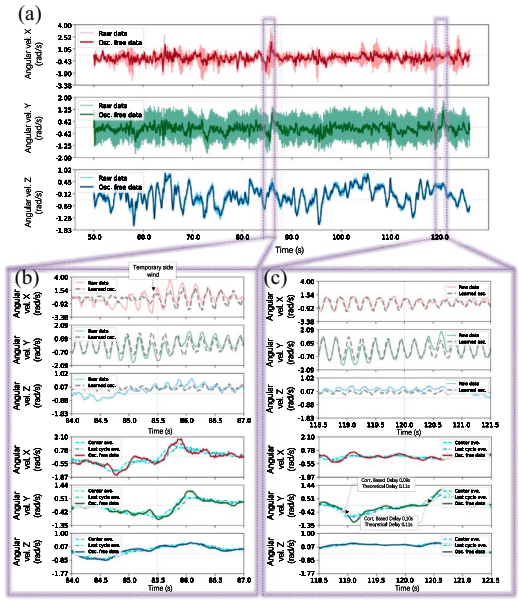


Fig. 9. Angular velocity oscillation learning in real flapping-wing flight. Theoretical delays are computed based on the estimated oscillation frequencies, while correlation-based delays are also provided for comparison. Signal labels are consistent with those defined in Fig. 7. "Last cycle ave." represents angular velocity signals averaged over the most recent flapping cycle. (a) Raw angular velocity signals and oscillation-free data over a 75-second flight segment. (b) A flight interval during which side-wind disturbances are present. (c) A flight interval characterized by rapid rotational motion without side-wind disturbances.

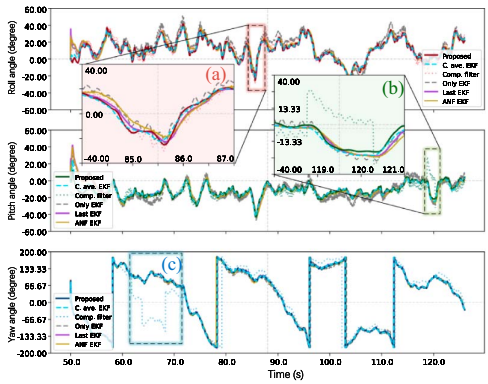


Fig. 10. Comparison of attitude estimation algorithms. In the legends, "C. ave. EKF" denotes an EKF using centered cycle-averaged (non-causal) measurements, serving as an approximate ground truth. The "Complementary filter" (Comp. filter) in PX4, a simplified Mahony filter, updates attitude estimates based on predefined triggers, leading to the observed switching behavior. "Only EKF" applies raw sensor measurements directly to the EKF, while "Last EKF" uses last-cycle-averaged filtered measurements. "ANF-EKF" preprocesses sensor signals using the proposed ANF before EKF estimation. (a)–(c) Highlight regions where the proposed method demonstrates clear performance advantages over the baseline approaches.

B. Attitude Estimation

The oscillation-free data obtained in last section were fed into the EKF attitude estimation algorithm mentioned previously. The obtained results are explicitly shown in Fig. 10. The estimation process initiates when oscillation-free data becomes accessible, and the result from the developed method gradually convergent to the centered cycle-average EKF results, which approximates the ground truth. In Fig. 10(a), the roll curve generated by the proposed method more closely aligns with the approximated ground truth (C. ave. EKF). Furthermore, severe oscillations result in considerable deterioration of estimation performance in methods lacking preprocessing (Comp. filter and Only EKF), as shown in both Fig. 10(b) and (c). This

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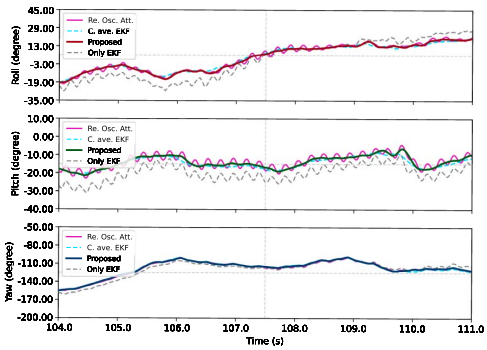


Fig. 11. Reconstruction of the oscillating attitude from the oscillation-free signals and the online learned pattern.

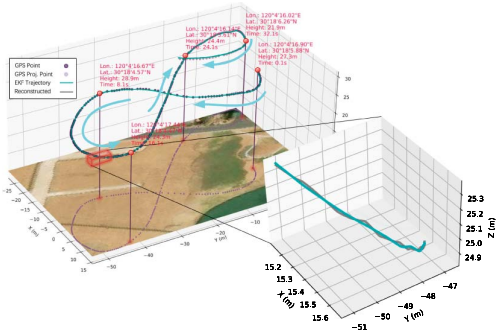


Fig. 12. Internal-model-based state estimation: the GPS location data are highlighted in violet, with detailed information presented every 8 s. These data points are overlaid on a flat satellite map, depicted in lavender. Measurements along the X and Y axes are derived from the GPS coordinates using the ellipsoidal model. The estimated trajectory is represented in teal, and the reconstructed one in gray.

degradation is effectively alleviated by the proposed approach. Moreover, as previously observed in Fig. 9, delays emerge in last cycle averaging based EKF. Similar delays can also be observed in ANF EKF, where parameters have already been optimized to achieve optimal performance.

Since the periodic patterns are learned online, we can reconstruct the oscillating attitude from the oscillation-free estimation. The obtained results are demonstrated in Fig. 11. First, the reconstructed curve fluctuates around the oscillation-free estimation with an amplitude equivalent to that of the only EKF estimation. Second, the periodic oscillation generated by only EKF estimation display a noticeable bias from the approximated ground truth. In conclusion, the proposed method effectively reconstructs the original oscillating attitude without bias, yielding a precise estimate of the current FWAV states, which can enhance future whole-body planning tasks.

C. Translational State Estimation

The internal-model-based EKF is developed to estimate slow-varying states, which include position, velocity and orientation, utilizing oscillation-free measurements obtained previously. In order to complete the internal model, we incorporate oscillation data into $C_{L,\mathrm{osc}}$, q_{osc} and $\sin(\psi_q - \psi_L)$ as presented in (18b) and (18c). The internal model outlined in (19) functions as predictive models on velocity. Furthermore, position and velocity measured by RTK play the observation roles. The EKF update rate is 200 HZ.

To validate the effectiveness of our internal-model-based state estimation approach, we perform this state estimation on a circle trajectory. As illustrated in Fig. 12, the estimated trajectory possesses higher resolution and aligns closely with that measured by RTK positioning. Using the obtained trajectory, we successfully reconstruct the oscillating original state by integrating the acceleration pattern.

D. Comparison Analysis

In FWAV researches, the approach to attitude estimation commonly varies in the preprocessing stage. Consequently, online methods can be classified into three primary categories: those based on cycle averaging [12], [43], those utilizing adaptive filters [15], and those relied on models [4], [11].

As shown in Table I, the proposed method fulfills all criteria, making it versatile and robust. It works without relying on angle data or prior models, adapts to both environmental and internal changes, estimates both states and oscillations, and maintains stable control feedback without delay or oscillations.

The absence of an encoder in our FWAV precludes the use of model-based strategies. Consequently, we provide a comparative analysis focusing on the cycle-averaging technique, the adaptive filter method, and the proposed approach. Meanwhile, the conventional EKF-based attitude estimation, as implemented previously, is retained for each method. The proposed method improves attitude estimation over the strongest baseline, i.e., Last EKF (see Table II), reducing Att. RMSE/MAE by about 24% in Case 1, 17% in Case 2, and 7% in Case 3, demonstrating consistent gains across all cases while also outperforming the baseline in other metrics.

E. Adaptability and Robustness Analysis

In order to validate the adaptability to flapping frequency varying, additional experiment was conducted. Fig. 13 reveals that, despite substantial variation in the flapping-wing frequency around $t \in [170, 185]$ s, the proposed method produces roll, pitch, and yaw estimates that closely match those of the centered cycle-average EKF. The estimated frequency adapts with bounded uncertainty, and the attitude trajectories remain virtually identical throughout the transition, demonstrating that the proposed estimator maintains robust and stable performance under frequency-dependent scenario.

The internal translational state estimation achieves robust performance in both internal and oscillation-source estimation across varying frequencies. The internal-model-based EKF produces smoother results than the conventional EKF. The root mean square errors (RMSEs) and mean absolute error (MAE) of velocity comparisons between the two EKFs during two cases are illustrated in Table III. Meanwhile, in Case 2, RMSE of position is 0.049 m and MAE in the xy plane is 0.046 and 0.004 m in Z direction, and in Case 3, RMSE of position is 0.058 m and MAE in the xy plane is 0.048 and 0.028 m in

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TABLE I
QUALITATIVE COMPARISON OF FWAV ATTITUDE ESTIMATION METHODS CONSIDERING PERIODIC OSCILLATION

Methods	Independence From Exact Angle Info.	Environmental and Self Adaptation	Internal state & Oscillation Est.	No theoretical Delay	Theoretical Oscillation Free	Independence from Prior Knowledge
Cycle-averaging	✓	✓	✗	✗	✓	✓
With adaptive filter	✓	✓	✗	✗	✗	✓
Model-based	✗	✗	✓	✓	✓	✗
Proposed	✓	✓	✓	✓	✓	✓

Note: ✓ indicates the property is satisfied; ✗ not satisfied. Six criteria for practical applications are: Angle independence: Avoid reliance on wing angle measurements from encoders. Environmental & self adaptation: Handle turbulence, wind, battery decline, actuator wear, and misalignments via agile GPR online learning. Oscillation-aware state estimation: Capture internal states for planning and control, and estimate oscillations for better state awareness and flight status detection. No theoretical delay: Prevent sensing or control delays that compromise stability in flapping dynamics. Oscillation-free feedback: Suppress flapping-induced oscillations that distort feedback and mislead planners or controllers. Model-free adaptability: Avoid reliance on detailed models or prior data to enable flexible, plug-and-play operation.

TABLE II
QUANTITATIVE COMPARISON OF FWAV ATTITUDE ESTIMATION

Metrics	Att. RMSE (rad)	Att. MAE (rad)	Acc. RMSE (m/s^2)	Acc. MAE (m/s^2)	Ang. v. RMSE (rad/s)	Ang. v. MAE (rad/s)
C1 Pro.	**0.035**	**0.031**	**1.661**	**1.424**	**0.243**	**0.207**
C1 Only	0.180	0.170	11.375	10.991	1.132	1.040
C1 Last	0.047	0.041	1.699	1.456	0.246	0.213
C1 ANF	0.079	0.071	7.217	6.449	0.640	0.539
C2 Pro.	**0.045**	**0.039**	2.792	2.392	**0.315**	**0.276**
C2 Only	0.319	0.233	11.167	10.686	1.348	1.239
C2 Last	0.055	0.047	**2.680**	**2.300**	0.325	0.285
C2 ANF	0.126	0.091	6.764	5.879	0.520	0.455
C3 Pro.	**0.029**	**0.026**	2.700	2.361	**0.237**	**0.196**
C3 Only	0.116	0.105	11.457	10.995	1.376	1.282
C3 Last	0.032	0.028	**2.499**	**2.172**	0.242	0.203
C3 ANF	0.042	0.038	6.673	5.801	0.561	0.496

Note: Optimal performances are highlighted in **bold**. The term RMSE means Root Mean Square Error, whereas MAE denotes Mean Absolute Error. Att. Acc. and Ang. v are abbreviations for attitude, acceleration, and angular velocity, respectively. C$\star$ stands for Case$\star$. And Pro., Only, Last, and ANF are defined identical to those in Fig. 10).

Z direction. As for periodic pattern recognition, the internal-model-based EKF preserves consistent patterns across frequencies, whereas the conventional EKF, limited by low-frequency RTK updates and frequency-dependent corrections, becomes less consistent at higher frequencies. Fig. 14 compares internal state and periodic pattern estimates of velocity along Z_I, where oscillation pattern is most obvious by EKFs at 200, 100, and 50 Hz. Smoother estimation and consistent recognition provide reliable AoA estimation and other details for aerodynamics-concerned flight control in future works.

V. CONCLUSION

This article introduces a novel learning-based framework for state estimation in avian-inspired FWAVs. The method leverages low-cost MARG sensors and a divide-and-conquer strategy grounded in cycle-averaged aerodynamics to separate high-frequency oscillatory disturbances from slow-varying dynamics. Experimental evaluations show that online periodic pattern learning—implemented via FFT, cross-correlation, and real-time clustering—effectively extracts oscillation-free signals, leading to notable improvements in attitude estimation accuracy. When incorporated into an internal-model-based EKF,

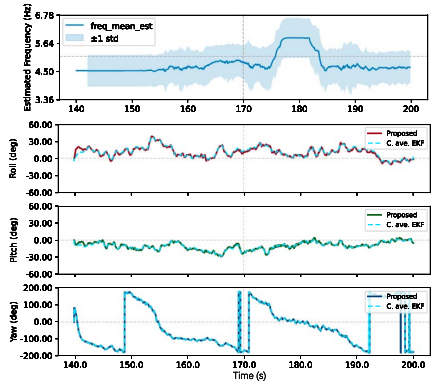


Fig. 13. Attitude estimation performance when the flapping-wing frequency undergoes significant variation.

TABLE III
QUANTITATIVE COMPARISON OF INTERNAL VELOCITY ESTIMATION

Metrics	Vel. RMSE (m/s)	Vel xy. MAE (m/s)	Vel z. MAE (m/s)
C2 Int. EKF	**0.048**	**0.019**	**0.028**
C2 Conv. EKF	0.053	0.024	0.032
C3 Int. EKF	**0.027**	**0.020**	**0.012**
C3 Conv. EKF	0.048	0.031	0.021

Note: Optimal performances are highlighted in **bold**. RMSE means Root Mean Square Error, and MAE denotes Mean Absolute Error. Vel. is abbreviations for velocity. The terms Vel. xy and Vel. z denote the horizontal and vertical velocities, respectively. Definitions of Int.EKF and Conv.EKF all follows Fig. 14 of internal estimations. $C_\star$ stands for Case $\star$. The omission of C_1 is attributed to strong winds.

these improved measurements yield state estimates closely aligned with high-precision positioning data, thereby enhancing flight stability and facilitates maneuverability in challenging environments.

Despite its demonstrated effectiveness, the method faces several challenges, including sensitivity to sensor noise and phase drift, computational overhead from online learning, and reduced robustness under abrupt maneuvers or changing environmental conditions. Its reliance on data-driven periodic modeling

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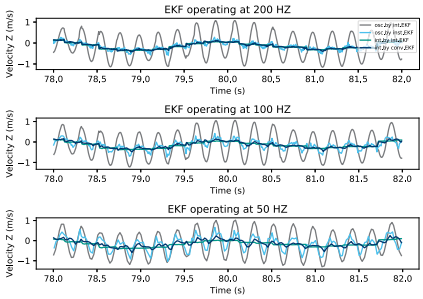


Fig. 14. Internal estimations of velocity along Z_I by conventional EKF uses online learned acceleration as inputs of prediction denoted as Int.by conv. EKF becomes less smooth as changing update frequency decreases, while EKF uses translational internal models as inputs of prediction denoted as Int.by int. EKF retains smooth estimations. State reconstruction between internal-model-based approach denoted as osc.by int.EKF and instantaneous-model-based EKF denoted as osc. by inst. EKF for reconstruction oscillation original velocity signal along Z_I. Albeit with the changing update frequency, reconstructed oscillating states denoted by gray lines retains stable pattern. Meanwhile, periodic pattern from instantaneous-model-based EKF denoted by blue lines fail to preserve its consistency.

also limits generalization to different morphologies or flight regimes. These challenges can be mitigated by combining robust multisensor fusion, lightweight learning algorithms, simplified physics-based models, and adaptive phase tracking to improve efficiency, resilience, and generalization. Future work will focus on addressing these issues to further advance the applicability of the proposed framework.

REFERENCES

[1] W. Shyy, M. Berg, and D. Ljungqvist, "Flapping and flexible wings for biological and micro air vehicles," *Prog. Aerosp. Sci.*, vol. 35, no. 5, pp. 455–505, Jul. 1999.

[2] J. Gerdes et al., "Robo raven: A flapping-wing air vehicle with highly compliant and independently controlled wings," *Soft Robot.*, vol. 1, no. 4, pp. 275–288, Dec. 2014.

[3] R. Zufferey et al., "Design of the high-payload flapping wing robot e-flap," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3097–3104, Apr. 2021.

[4] W. Xu, E. Pan, J. Liu, Y. Li, and H. Yuan, "Flight control of a large-scale flapping-wing flying robotic bird: System development and flight experiment," *Chin. J. Aeronaut.*, vol. 35, no. 2, pp. 235–249, Feb. 2022.

[5] A. Chen, B. Song, Z. Wang, D. Xue, and K. Liu, "A novel actuation strategy for an agile bioinspired FWAV performing a morphing-coupled wingbeat pattern," *IEEE Trans. Robot.*, vol. 39, no. 1, pp. 452–469, Aug. 2022.

[6] G. Liu, S. Wang, and W. Xu, "Flying state sensing and estimation method of large-scale bionic flapping wing flying robot," *Actuators*, vol. 11, no. 8, Jul. 2022, Art. no. 213.

[7] X. Wu, W. He, Q. Wang, T. Meng, X. He, and Q. Fu, "A long-endurance flapping-wing robot based on mass distribution and energy consumption method," *IEEE Trans. Ind. Electron.*, vol. 70, no. 8, pp. 8215–8224, Aug. 2023.

[8] N. Trawny and S. I. Roumeliotis, "Indirect Kalman filter for 3D attitude estimation," *Dept. Comput. Sci. Eng., Univ. Minnesota, Tech. Rep.*, vol. 2, no. 2005, p. 2005, 2005.

[9] R. Mahony, T. Hamel, and J.-M. Pflimlin, "Nonlinear complementary filters on the special orthogonal group," *IEEE Trans. Autom. Control*, vol. 53, no. 5, pp. 1203–1218, Aug. 2008.

[10] S. O. H. Madgwick, "An efficient orientation filter for inertial and inertial/magnetic sensor arrays," *Univ. Bristol, Bristol, U.K., Tech. Rep.*, 2010.

[11] Z. Tu, F. Fei, Y. Yang, J. Zhang, and X. Deng, "Realtime on-board attitude estimation of high-frequency flapping wing MAVs under large instantaneous oscillation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Piscataway, NJ, USA: IEEE Press, 2018, pp. 6806–6811.

[12] J. Verboom, S. Tijmons, C. De Wagter, B. Remes, R. Babuska, and G. C. de Croon, "Attitude and altitude estimation and control on board a flapping wing micro air vehicle," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Piscataway, NJ, USA: IEEE Press, 2015, pp. 5846–5851.

[13] S. Armanini, M. Karásek, G. De Croon, and C. De Visser, "On-board/offboard sensor fusion for high-fidelity flapping-wing robot flight data," *J. Guid. Control Dyn.*, vol. 40, no. 8, pp. 2121–2132, Aug. 2017.

[14] J. A. Grauer and J. E. Hubbard, Jr, "Inertial measurements from flight data of a flapping-wing ornithopter," *J. Guid. Control Dyn.*, vol. 32, no. 1, pp. 326–331, Jan. 2009.

[15] S. M. Mousavi and S. H. Pourtakdoust, "Improved neural adaptive control for nonlinear oscillatory dynamic of flapping wings," *J. Guid., Control, Dyn.*, vol. 46, no. 1, pp. 97–113, 2023.

[16] B. Corban, M. Bauerheim, and T. Jardin, "Discovering optimal flapping wing kinematics using active deep learning," *J. Fluid Mech.*, vol. 974, no. A54, pp. 1–28, Nov. 2023.

[17] T. Kim et al., "Wing-strain-based flight control of flapping-wing drones through reinforcement learning," *Nature Mach. Intell.*, vol. 6, no. 9, pp. 992–1005, Sep. 2024.

[18] J. Cai, V. Sangli, M. Kim, and K. Sreenath, "Learning-based trajectory tracking for bird-inspired flapping-wing robots," in *Proc. Am. Control Conf. (ACC)*, Piscataway, NJ, USA: IEEE Press, 2025, pp. 430–437.

[19] P. Chirarattananon, K. Y. Ma, and R. J. Wood, "Adaptive control of a millimeter-scale flapping-wing robot," *Bioinspir. Biomim.*, vol. 9, no. 2, May 2014, Art. no. 025004.

[20] Z. Tu, F. Fei, J. Zhang, and X. Deng, "An at-scale tailless flapping-wing hummingbird robot. I. Design, optimization, and experimental validation," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1511–1525, May 2020.

[21] C. Qian, Y. Fang, F. Jia, J. Yan, Y. Liang, and T. Li, "Toward practical autonomous flight simulation for flapping wing biomimetic robots with experimental validation," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 5084–5095, 2025.

[22] Y. Lu, X. Li, Y. Bai, and R. Sun, "Low-complexity performance enhancing flapping control for two-section FAVs: Design, analysis, and experiments," *IEEE Trans. Ind. Electron.*, vol. 72, no. 6, pp. 6362–6371, Jun. 2025.

[23] S. P. Sane and M. H. Dickinson, "The aerodynamic effects of wing rotation and a revised quasi-steady model of flapping flight," *J. Exp. Biol.*, vol. 205, no. 8, pp. 1087–1096, Apr. 2002.

[24] R. McGill, N.-s. P. Hyun, and R. J. Wood, "Modeling and control of flapping-wing micro-aerial vehicles with harmonic sinusoids," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 746–753, Dec. 2021.

[25] W. Siqi, S. Bifeng, C. Ang, F. Qiang, and C. Jin, "Modeling and flapping vibration suppression of a novel tailless flapping wing micro air vehicle," *Chin. J. Aeronaut.*, vol. 35, no. 3, pp. 309–328, Mar. 2022.

[26] E. Sanchez-Laulhe, R. Fernandez-Feria, and A. Ollero, "Simplified model for forward-flight transitions of a bio-inspired unmanned aerial vehicle," *Aerospace*, vol. 9, no. 10, Oct. 2022, Art. no. 617.

[27] L. Wang, W. Jiang, L. Zhao, and Z. Jiao, "Modeling and hover flight control of a micromechanical flapping-wing aircraft inspired by wing–tail interaction," *IEEE/ASME Trans. Mechatronics*, vol. 28, no. 6, pp. 3132–3142, Mar. 2023.

[28] D. R. Warrick, M. Bundle, and K. Dial, "Bird maneuvering flight: Blurred bodies, clear heads," *Integr. Comp. Biol.*, vol. 42, no. 1, pp. 141–148, 2002.

[29] N. Durrande, J. Hensman, M. Rattray, and N. D. Lawrence, "Detecting periodicities with gaussian processes," *PeerJ Comput. Sci.*, vol. 2, Apr. 2016, Art. no. e50.

[30] J. Quinonero-Candela and C. E. Rasmussen, "A unifying view of sparse approximate gaussian process regression," *J. Mach. Learn. Res.*, vol. 6, pp. 1939–1959, Dec. 2005.

[31] M. Puschel and J. M. Moura, "Algebraic signal processing theory: Cooley–Tukey type algorithms for DCTS and DSTS," *IEEE Trans. Signal Process.*, vol. 56, no. 4, pp. 1502–1521, Apr. 2008.

[32] N. R. Lomb, "Least-squares frequency analysis of unequally spaced data," *Astrophys. Space Sci.*, vol. 39, no. 2, pp. 447–462, 1976.

[33] J. D. Scargle, "Studies in astronomical time series analysis. II-statistical aspects of spectral analysis of unevenly spaced data," *Astrophys. J.*, vol. 263, pp. 835–853, Dec. 1982.

[34] K. Hornik, I. Feinerer, M. Kober, and C. Buchta, "Spherical k-means clustering," *J. Statist. Softw.*, vol. 50, no. 1, pp. 1–22, Sep. 2012.

[35] A. M. Sabatini, "Kalman-filter-based orientation determination using inertial/magnetic sensors: Observability analysis and performance evaluation," *Sensors*, vol. 11, no. 10, pp. 9182–9206, Sep. 2011.

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[36] D. M. Wolpert, Z. Ghahramani, and M. I. Jordan, "An internal model for sensorimotor integration," *Science*, vol. 269, no. 5232, pp. 1880–1882, 1995.
[37] J. Alaminos-Quesada, "Limit of the two-dimensional linear potential theories on the propulsion of a flapping airfoil in forward flight in terms of the Reynolds and Strouhal number," *Phys. Rev. Fluids*, vol. 6, no. 12, p. 123101, 2021.
[38] F. Rodríguez, J.-M. Díaz-Báñez, E. Sanchez-Laulhe, J. Capitán, and A. Ollero, "Kinodynamic planning for an energy-efficient autonomous ornithopter," *Comput. Ind. Eng.*, vol. 163, Jan. 2022, Art. no. 107814.
[39] R. Fernandez-Feria, "Linearized propulsion theory of flapping airfoils revisited," *Phys. Rev. Fluids*, vol. 1, no. 8, Dec. 2016, Art. no. 084502.
[40] Y. Lee, K.-B. Lua, T. Lim, and K. Yeo, "A quasi-steady aerodynamic model for flapping flight with improved adaptability," *Bioinspiration Biomimetics*, vol. 11, no. 3, 2016, Art. no. 036005.
[41] R. W. Beard and T. W. McLain, *Small Unmanned Aircraft: Theory and Practice*. Princeton, NJ, USA: Princeton Univ. Press, 2012.
[42] P. Duhamel and M. Vetterli, "Fast Fourier transforms: A tutorial review and a state of the art," *Signal Process.*, vol. 19, no. 4, pp. 259–299, 1990.
[43] G. C. De Croon, J. J. Dupeyroux, C. De Wagter, A. Chatterjee, D. A. Olejnik, and F. Ruffier, "Accommodating unobservability to control flight attitude with optic flow," *Nature*, vol. 610, no. 7932, pp. 485–490, 2022.



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